



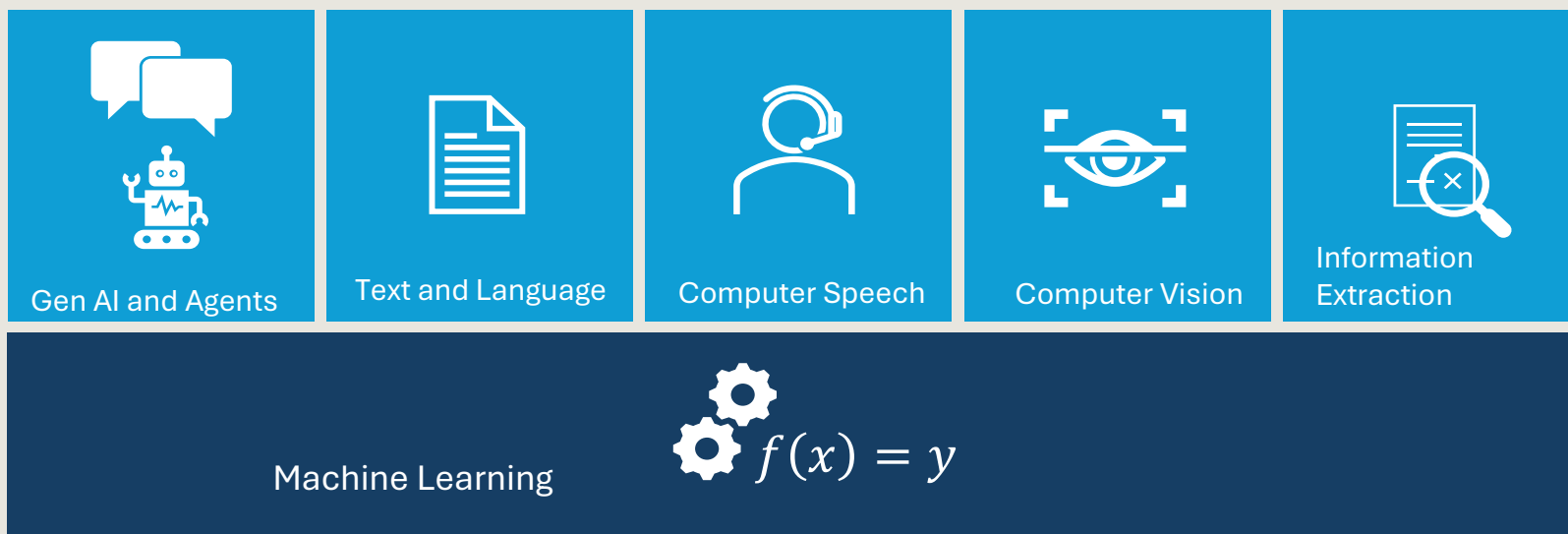
Extract insights from visual data in Azure

<https://aka.ms/mslearn-visual-insights>

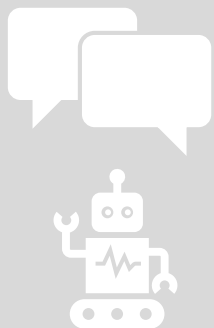


Agenda

1. Develop generative AI apps in Azure
2. Develop AI agents in Azure
3. Develop natural language solutions in Azure
4. Extract insights from visual data on Azure



Our focus today



Generative AI and Agents



Text and Language



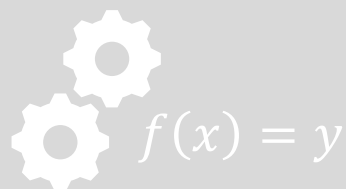
Computer Speech



Computer Vision



Information Extraction



Machine Learning

Develop a vision-enabled generative AI application

<https://aka.ms/mslearn-gen-ai-vision>

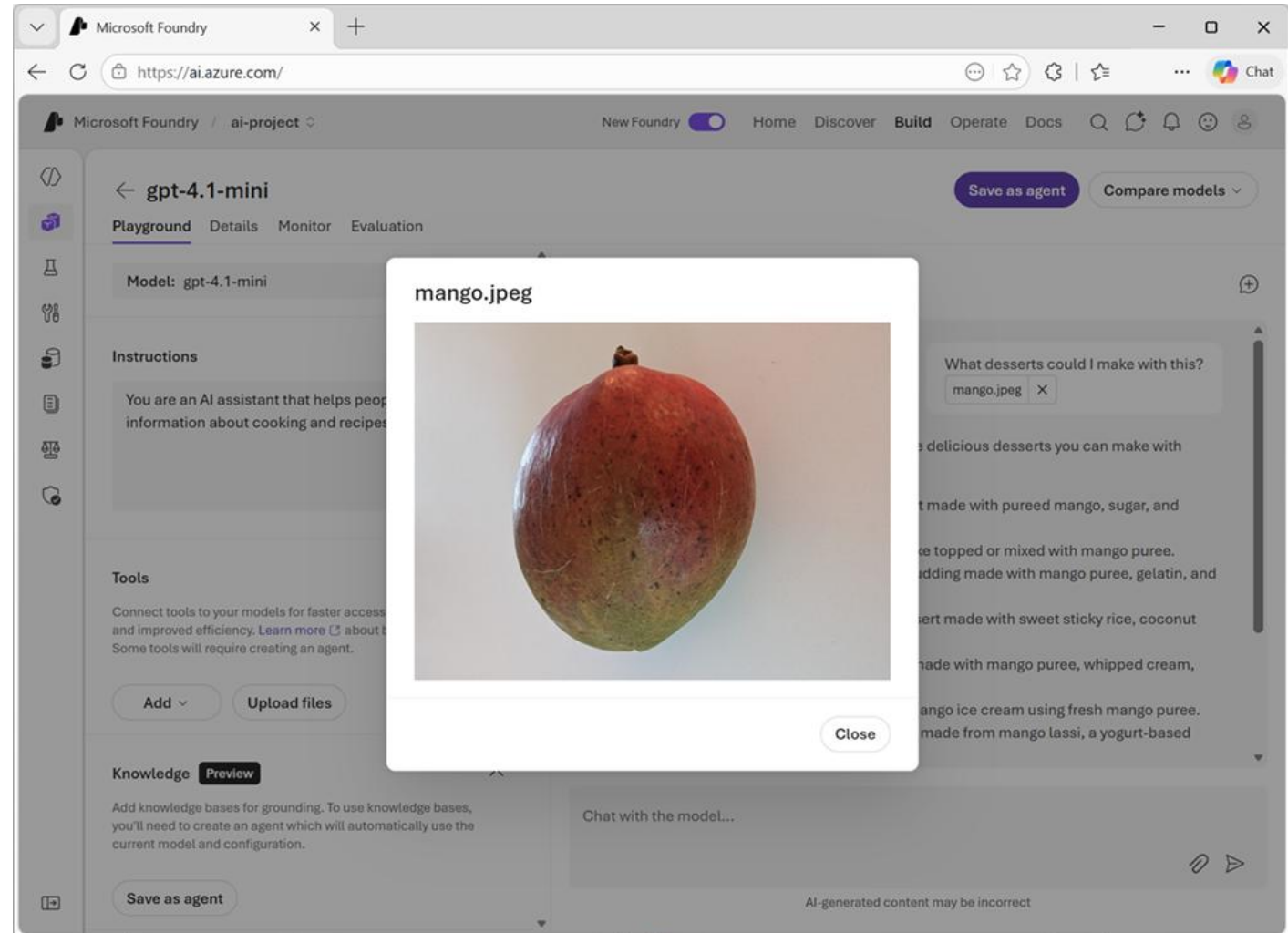


Using a vision-capable model

Multimodal models for text and image data:

- Text prompts
- Image prompts
- Multimodal prompts

Typically, multimodal prompts for vision-based input include multiple parts (for example, a text part and an image part)



Compare APIs for image-based prompts



ChatCompletions API

1 LOAD
IMAGE

```
image_path = Path("fruit.jpeg")
image_format = "jpeg"
with open(image_path, "rb") as image_file:
    image_data = base64.b64encode(
        image_file.read()).decode("utf-8")
```

2 CREATE
URL

```
data_url = f"data:image/{image_format};base64,{image_data}"
```

3 CALL
API

```
response = client.chat.completions.create(
    model="Phi-4-multimodal-instruct",
    messages=[
        {"role": "system", "content": "You are a chef."},
        {"role": "user", "content": [
            {"type": "text", "text": "What can I make?"},
            {"type": "image_url", "image_url": {"url": data_url}}
        ]}
    ]
)
```

4 GET
OUTPUT

```
print(response.choices[0].message.content)
```



Responses API

```
image_path = Path("fruit.jpeg")
image_format = "jpeg"
with open(image_path, "rb") as image_file:
    image_data = base64.b64encode(
        image_file.read()).decode("utf-8")
```

```
data_url = f"data:image/{image_format};base64,{image_data}"
```

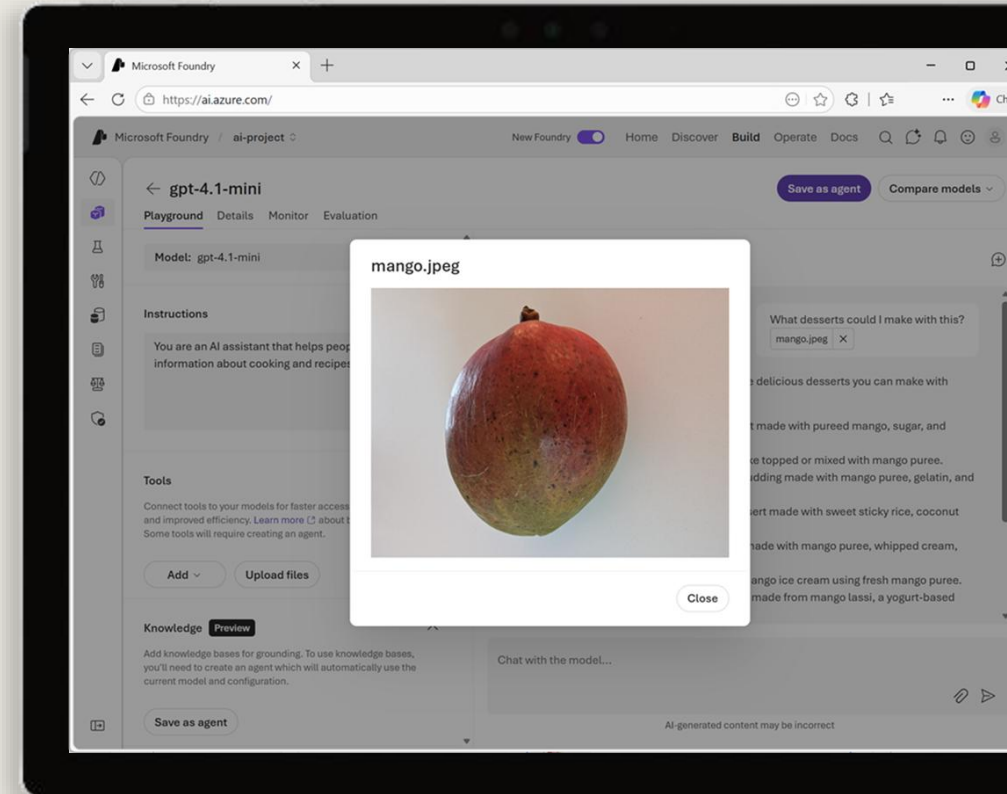
```
response = client.responses.create(
    model="gpt-4.1", input=[
        {"role": "developer", "content": "You are a chef."},
        {"role": "user", "content": [
            {"type": "input_text", "text": "What can I make?"},
            {"type": "input_image", "image_url": data_url}
        ]}
    ]
)
```

```
print(response.output_text)
```

Demo

Develop a vision-enabled chat app

In this demo, you'll see how to generate responses to prompts that include images. You'll develop an app that provides AI assistance with fresh produce in a grocery store by using Microsoft Foundry and the OpenAI SDK.



Knowledge check



1 Which kind of model can you use to respond to visual input?

- ☐ Only OpenAI GPT models
- ☐ Embedding models
- ☒ Multimodal models

2 How should you submit a prompt that asks a model to analyze an image?

- ☐ Submit one prompt with an image-based message followed by another prompt with a text-based message
- ☒ Submit a prompt that contains a multi-part user message, containing both text content and image content
- ☐ Submit the image as the system message and the instruction or question as the user message

3 How can you include an image in a message?

- ☒ As a URL or as binary data
- ☐ Only as a URL for an online image
- ☐ Only as binary data

Generating images and video

<https://aka.ms/mslearn-generate-images>

<https://aka.ms/mslearn-generate-video>



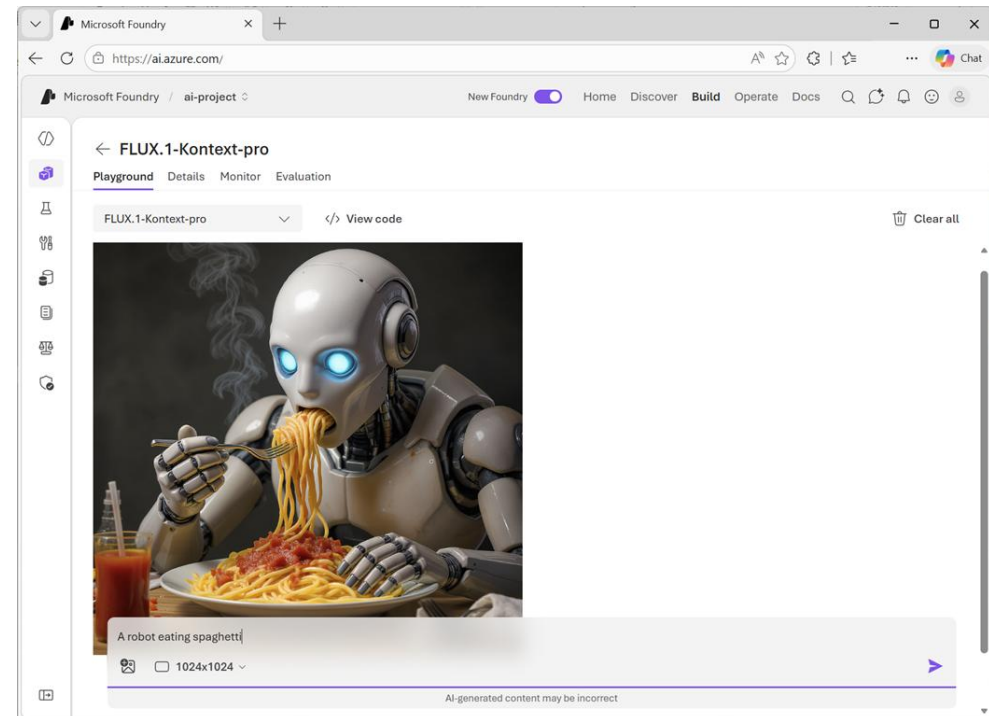
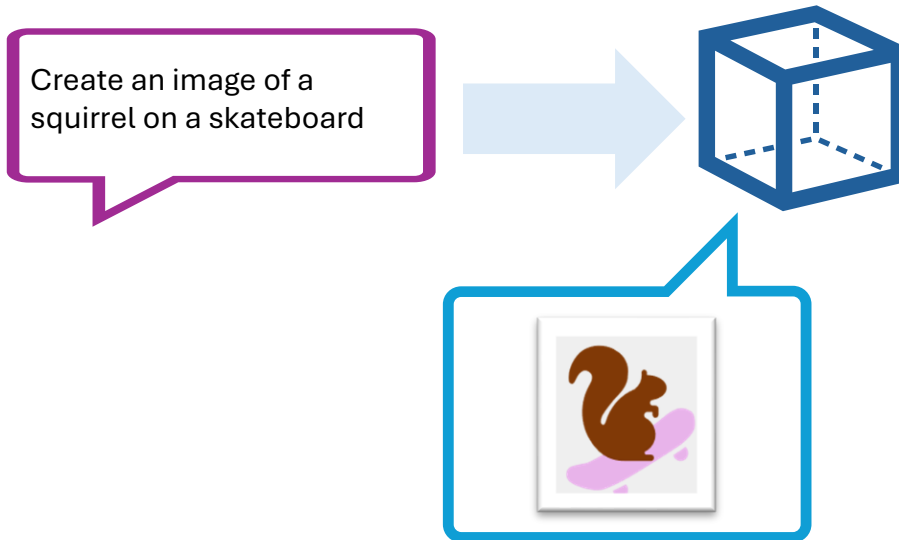
Using an image-generation model

Deploy an image-generation model:

- GPT-Image
- FLUX.1-Kontext-pro
- ...

Test the model in the playground

- Select an image size
- Attach a reference image
- (if supported and required)
- Enter a prompt describing the desired image

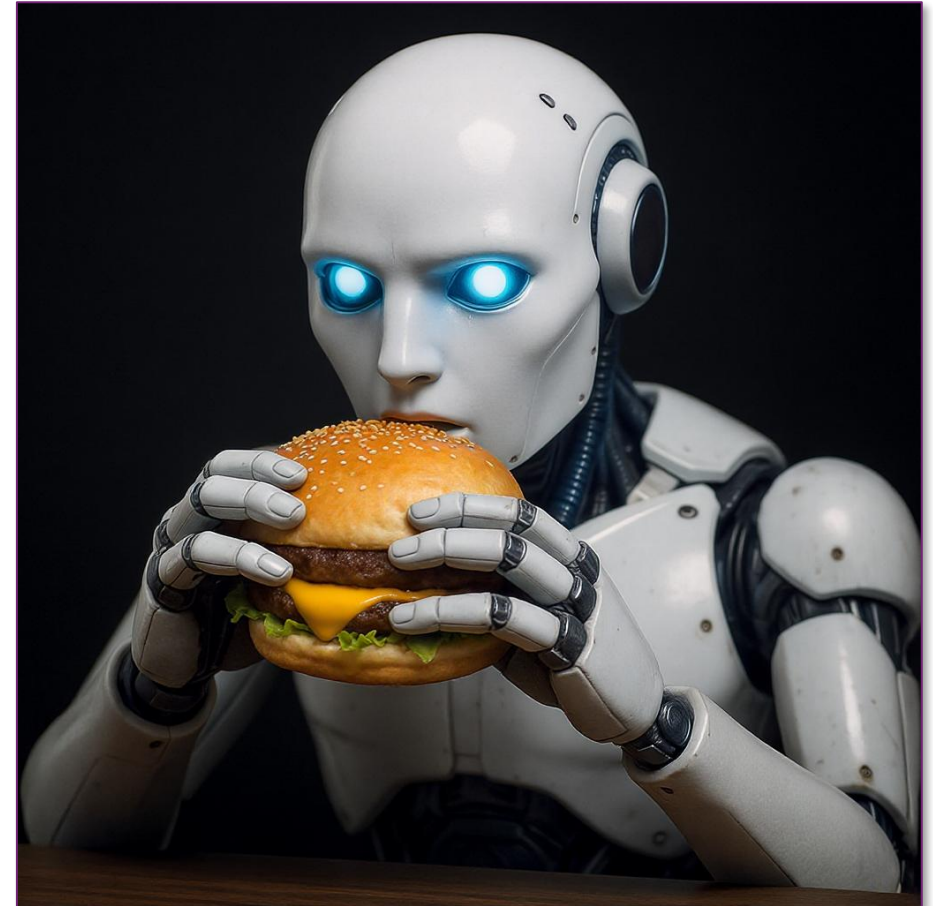


Generating an image with the OpenAI *Image* API

```
# Generate an image
img_results = client.images.generate(
    model="FLUX.1-Kontext-pro",
    prompt="A robot eating a cheeseburger.",
    n=1,
    size="1024x1024",
)

# Save the generated image
image_data = base64.b64decode(img_results.data[0].b64_json)

with open("image.png", "wb") as image_file:
    image_file.write(image_data)
```

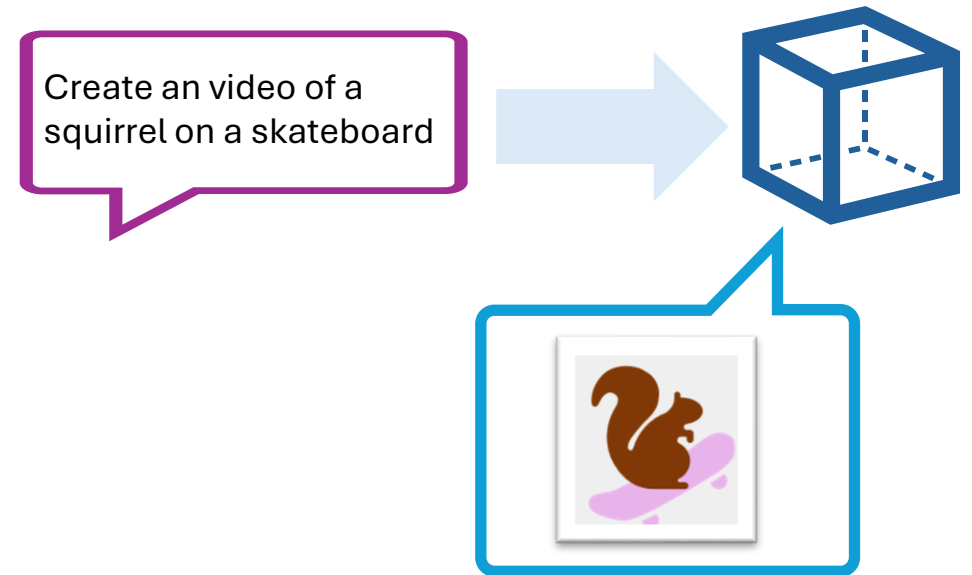


Video-generating models

Some generative AI models can create video scenes from text prompts.

For example, Sora 2 is a model that can create generate realistic scenes from text prompts, reference images, or by remixing existing videos.

You can deploy and use the Sora 2 model in Microsoft Foundry projects



Generate videos from prompts

For effective video prompts, describe a shot for a storyboard. Include details such as:

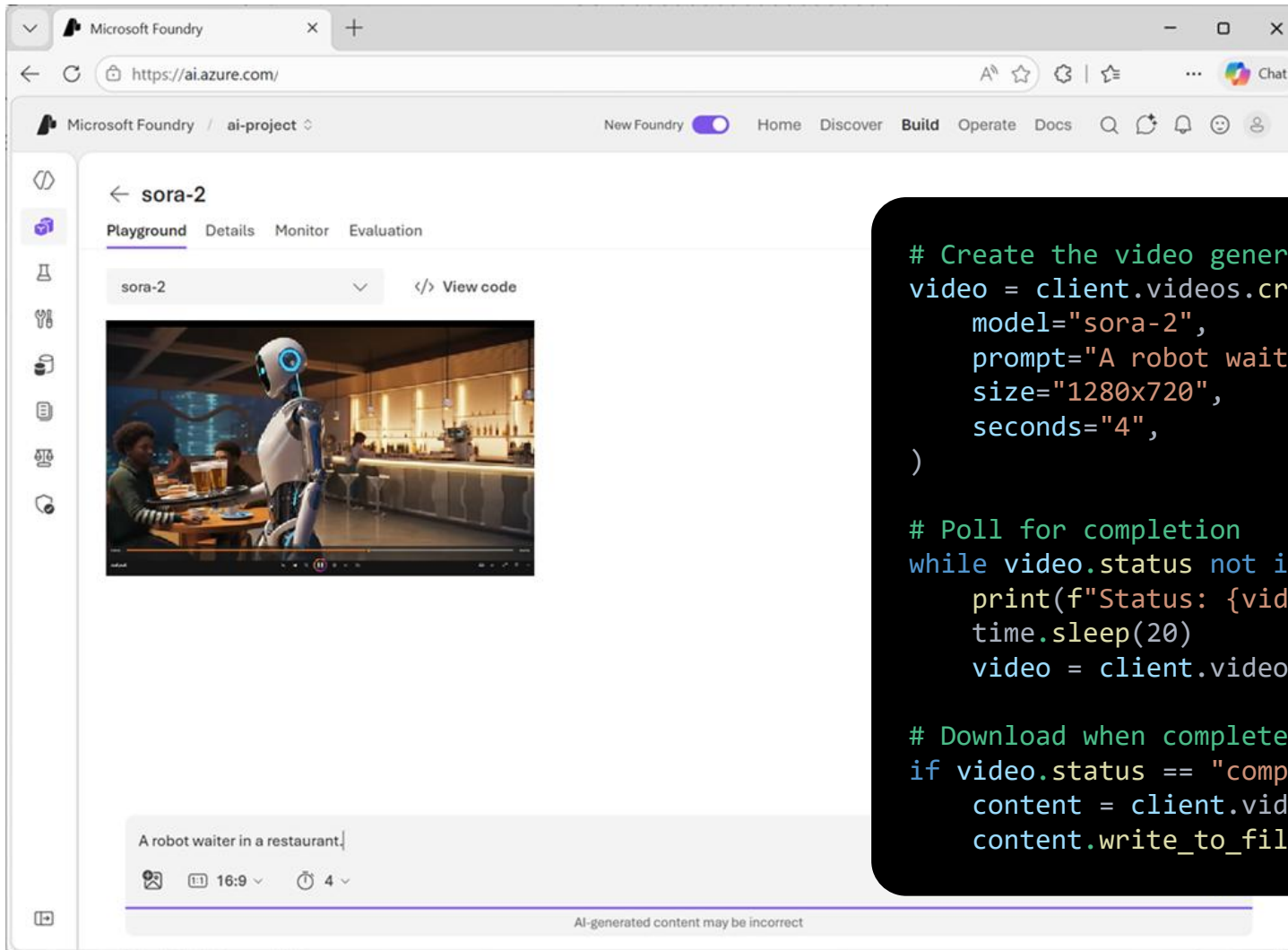
- Camera framing (wide, medium, close-up)
- Subject description with distinctive details
- Action described in small steps
- Lighting, color palette, and style



Example

In a 1980s sci-fi film style, a medium close-up shot at eye level captures a young woman standing at the edge of a rain-soaked rooftop. She takes two slow steps forward, pauses, and turns her head to look over her shoulder. Neon pink and blue light spills from billboards behind her. The color palette anchors are electric blue, hot pink, and steel gray.

Using a video-generation model



```
# Create the video generation job
video = client.videos.create(
    model="sora-2",
    prompt="A robot waiter in a restaurant.",
    size="1280x720",
    seconds="4",
)

# Poll for completion
while video.status not in ["completed", "failed", "cancelled"]:
    print(f"Status: {video.status}. ...")
    time.sleep(20)
    video = client.videos.retrieve(video.id)

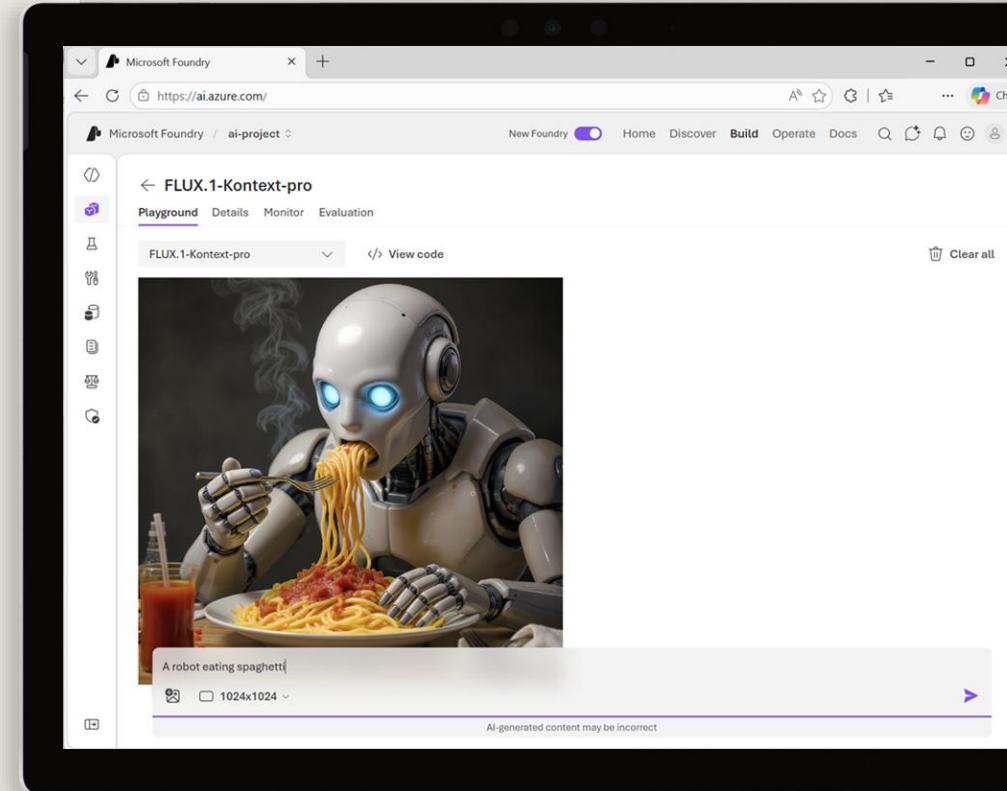
# Download when complete
if video.status == "completed":
    content = client.videos.download_content(video.id, variant="video")
    content.write_to_file("output.mp4")
```

Demo

Generate images with AI

In this demo, you'll see how to:

- Use an image-generation model in the playground
- Develop a client app that generates images



Knowledge check



1 You want to find a model in Microsoft Foundry to generate images. Which inference task should you filter by?

- ☒ Text to image
- ☐ Image to text
- ☐ Embeddings

2 Which OpenAI API can you use with image-generation models?

- ☐ Video
- ☒ Image
- ☐ Graphics

Analyze images with Content Understanding

<https://aka.ms/mslearn-image-content-understanding>



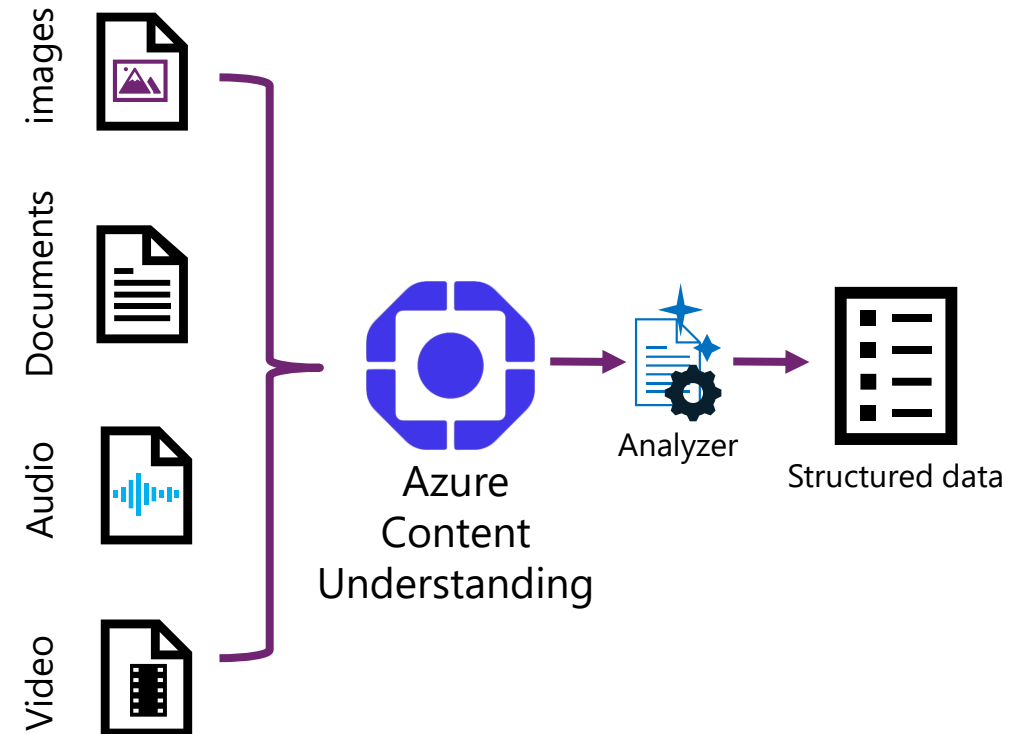
What is Azure Content Understanding in Foundry Tools?

Multimodal content analysis:

- Images
- Documents and forms
- Audio
- Video

AI-powered information extraction

- Pre-built and custom analyzers
- Single/Few-shot schema training



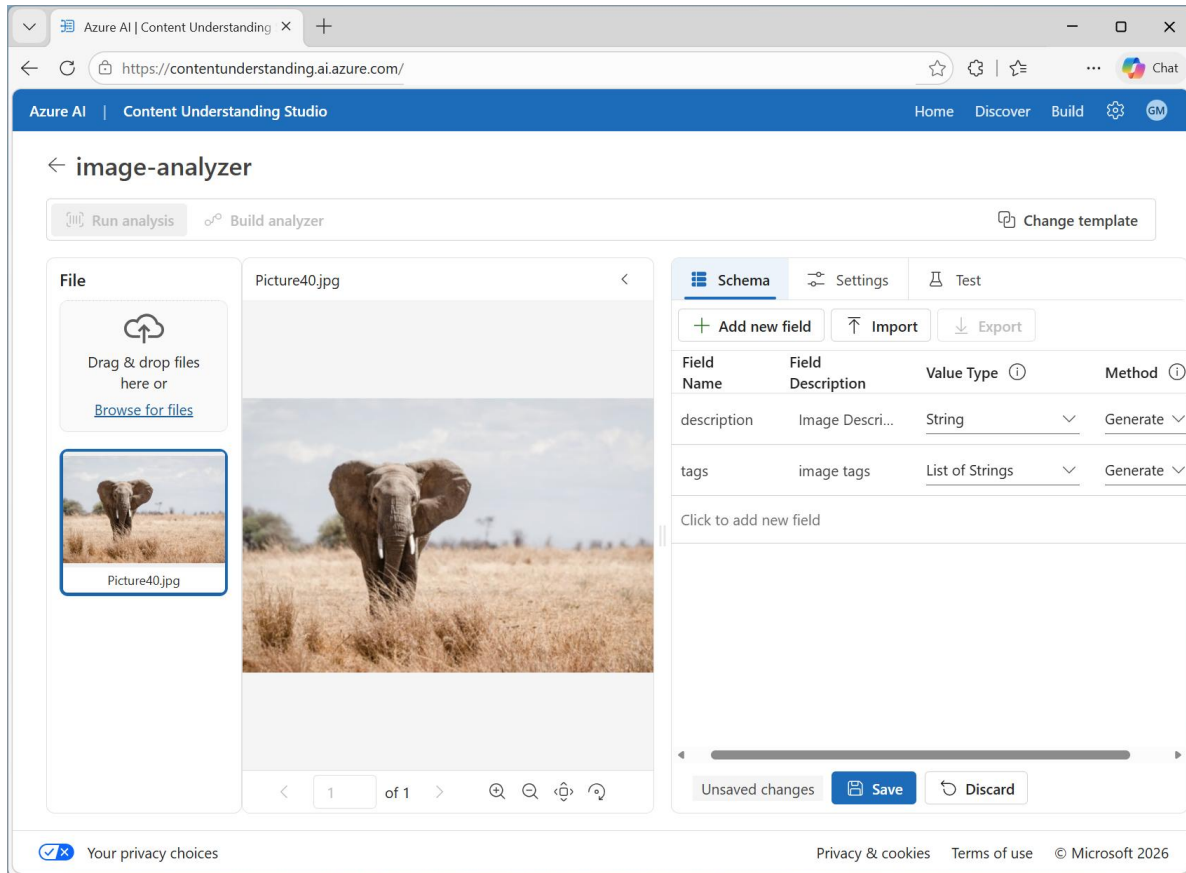
Analyzing images with Content Understanding



Analysis summary

The image shows a busy underground train station platform with a train passing by at high speed, creating a motion blur effect. Several people are walking along the platform, some appearing blurred due to motion. The station has a curved, arched ceiling with visible metal framework and lighting. A digital signboard above the platform indicates a train arrival in 3 minutes.

Analyzing images with Content Understanding



Use one or more training images to define a schema for a custom *analyzer*

```
# Get a Content Understanding client
client = ContentUnderstandingClient(
    endpoint={FOUNDRY_ENDPOINT},
    credential={KEY_OR_IDENTITY},
    api_version="2025-11-01"
)

# Submit an image file to the analyzer
with open("my_image.png", "rb") as f:
    file_bytes = f.read()

poller = client.begin_analyze(
    analyzer_id={ANALYSER_ID},
    inputs=[AnalysisInput(data=file_bytes)],
)

# Get results from poller
result: AnalysisResult = poller.result()

# Display results
result_str = json.dumps(result.as_dict(), indent=2)
print(result_str)
```

Use the Content Understanding API to run analyzers and generate results

Analysis results

Azure AI | Content Understanding Studio

Home Discover Build

image-analyzer

Run analysis Build analyzer Change template

File

image-1.png

Drag & drop files here or
[Browse for files](#)

image-1.png

Schema Settings **Test**

Fields Content Result

Description

A person is sitting on a purple couch with their legs crossed, using a laptop. There is an electric guitar resting on the couch beside them. The setting appears to be a cozy living room with plants in the foreground and a wooden coffee table with items on it. The person's face is blurred, and they are dressed casually in a grey t-shirt and blue jeans.

Tags (8)

- 1 living room
- 2 couch
- 3 laptop
- 4 electric guitar
- 5 plants
- 6 coffee table
- 7 casual

Check token usage in JSON (Result tab) and learn about pricing [here](#)

Home Discover Build

Change template

Schema Settings **Test**

Fields Content **Result**

JSON

```
4  "result": {
9    "contents": [
10   {
12     "markdown": "![image](pages/1)\n",
13     "fields": {
14       "Description": {
15         "type": "string",
16         "valueString": "A person is sitting on a
17       },
18       "Tags": {
19         "type": "array",
20         "valueArray": [
21           {
22             "type": "string",
23             "valueString": "living room"
24           },
25           {
26             "type": "string",
27             "valueString": "couch"
28           },
29           {
30             "type": "string",
31             "valueString": "laptop"
32           },
33           {
34             "type": "string",
35             "valueString": "electric guitar"
36           },
37           {
38             "type": "string",
```

Exercise

Analyze images with Azure Content Understanding

In this exercise, you'll:

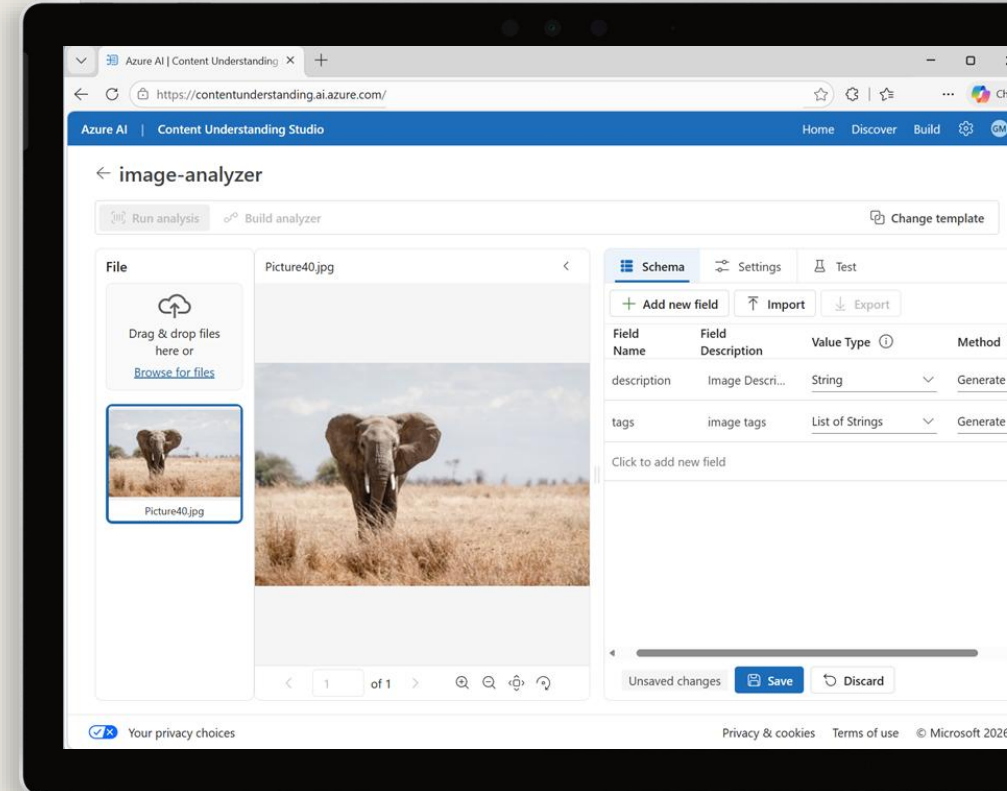
- Create an image analyzer in Azure Content Understanding
- Use the Content Understanding API to submit images for analysis

Start the exercise at:

20. Analyze images with Azure Content Understanding

🕒 1 h

0%



Knowledge check



1 You want to find a model in Microsoft Foundry to generate images. Which inference task should you filter by?

- ☒ Text to image
- ☐ Image to text
- ☐ Embeddings

2 Which OpenAI API can you use with image-generation models?

- ☐ Video
- ☒ Image
- ☐ Graphics

Analyze documents with Content Understanding

<https://aka.ms/mslearn-analyze-content-understanding>

<https://aka.ms/mslearn-ai-content-understanding-api>



Analyzing documents

Use Content Understanding to analyze documents and forms and retrieve specific field values

For example, extract key data values from an invoice to automate payment processing

Azure AI | Content Understanding Studio

Home Discover Build

Try prebuilt analyzer

Select a modality to begin. Then select the following options to specify the prebuilt analyzer.

Document Document fields [Learn more about prebuilt analyzer](#)

Drag & drop files here or [Browse for files](#)

document_fields.png

Run analysis Auto run analysis ☒

Level 10, Tower C, DLF Epitome Building No. 5, DLF Cyber City, Gurgaon 122002 01-12-2022 ₹ 6,54,321.00 01-03-2023

State Code: GSTIN: PLACE OF SUPPLY: Maharashtra

S.NO	DESCRIPTION	HSN/SAC	IGST	SGST	CGST	AMOUNT
1	Computer system, monitor and peripherals	850700	12%	6%	6%	₹ 6,00,000.00
1	Computer system, monitor and peripherals	850700	12%	6%	6%	₹ 1,00,000.00

Account Name: Sonoo Ltd SUBTOTAL ₹ 6,00,000.00
IFS CODE: MSFT00122344 IGST ₹ 1,08,000.00
Swift Code: ASULIN1111 SGST ₹ 0.00
Account Number: 2008170000000000 CGST ₹ 0.00

Fields Result Code

Hide missing fields ☒

- InvoiceNumber p.1 98.00% MSFT0011223344
- InvoiceDate p.1 98.00% 2022-12-31
- DueDate p.1 98.20% 2023-03-31
- InvoiceAmount p.1 46.60% 654321

Check token usage in JSON (Result tab) and learn about pricing [here](#)

✓ Your privacy choices

Privacy & cookies Terms of use © Microsoft 2026

Analyzing audio

Analyze audio to:

- Summarize conference calls
- Determine sentiment of recorded customer conversations
- Extract key data from telephone messages
- ...

The screenshot displays the 'Try prebuilt analyzer' interface in the Azure AI Content Understanding Studio. The top navigation bar includes 'Azure AI | Content Understanding Studio' and links for 'Home', 'Discover', 'Build', and a user profile icon. The main heading is '< Try prebuilt analyzer'. Below this, a instruction box says 'Select a modality to begin. Then select the following options to specify the prebuilt analyzer.' It features two dropdown menus: 'Audio' and 'Audio search', along with a link 'Learn more about prebuilt analyzer'. On the left, there's a file upload section with a cloud icon and text 'Drag & drop files here or [Browse for files](#)'. Below this is a sample audio file 'audio_analysis.mp3' with a play button icon. The central area contains a 'Run analysis' button, an 'Auto run analysis' toggle switch, and a progress bar showing '00:00' to '01:02'. Below the progress bar is a 'Content' section displaying a transcript of a customer service call. The transcript includes timestamps and speaker labels, such as '<v Speaker 1>Good afternoon, thank you for calling Contoso, how may I help?' and '<v Speaker 2>I am looking to transfer money to my checking account.' On the right, there's a 'Fields' tab with a 'Summary' section. The summary text reads: 'The conversation is a customer service call where Mary Smith inquires about transferring money from a matured \$1,000 certificate of deposit to her checking account. The agent from Contoso explains that the bank will handle the transfer without needing to contact the previous bank, requests her name and account number, and informs her that the transaction will take two...'. At the bottom of the summary, there's a note: 'Check token usage in JSON (Result tab) and learn about pricing [here](#).' The footer of the interface includes a privacy notice 'Your privacy choices' and links for 'Privacy & cookies', 'Terms of use', and '© Microsoft 2026'.

Analyzing video

Analyze video to:

- Extract key points from video conference recordings
- Summarize presentations
- Detect the presence of specific activity in security footage
- ...

The screenshot displays the Azure AI Content Understanding Studio interface. At the top, the navigation bar includes 'Azure AI | Content Understanding Studio', 'Home', 'Discover', 'Build', and a user profile icon. The main heading is 'Try prebuilt analyzer'. Below this, a section titled 'Select a modality to begin. Then select the following options to specify the prebuilt analyzer.' contains a 'Video' dropdown and a 'Video search' dropdown, with a link to 'Learn more about prebuilt analyzer'. On the left, there's a 'Drag & drop files here or Browse for files' area and a 'Sample' video player showing a flight simulator. The central video player shows a video titled 'flight_simulato... ch.mp4' with a 'Run analysis' button and an 'Auto run analysis' toggle. Below the video player is a 'Content' section with a transcript. On the right, a 'Fields' tab is active, showing a 'Summary' section with a detailed description of the video content, mentioning 'Flight Simulator logo' and 'Microsoft Azure AI branding'. At the bottom, there are links for 'Your privacy choices', 'Privacy & cookies', 'Terms of use', and '© Microsoft 2026'.

Azure AI | Content Understanding Studio

Home Discover Build

Try prebuilt analyzer

Select a modality to begin. Then select the following options to specify the prebuilt analyzer.

Video Video search [Learn more about prebuilt analyzer](#)

Drag & drop files here or [Browse for files](#)

Sample

flight_simulato... ch.mp4

Run analysis Auto run analysis ☒

00:00 00:43

Content

00:07.120 --> 00:13.320
<Speaker 2>To achieve that, we build a universal TTS model based on 3,000 hours of data.

00:13.440 --> 00:23.680
<Speaker 1>We actually accumulated tons of the data so that this

Fields Result Code

Shot 1 00:00

Summary

The video opens with a Flight Simulator logo alongside Microsoft Azure AI branding, followed by a discussion about neural text-to-speech (TTS) technology. A speaker explains the importance of having high-quality data, mentioning the creation of a universal TTS model trained on 3,000 hours of data to capture audio

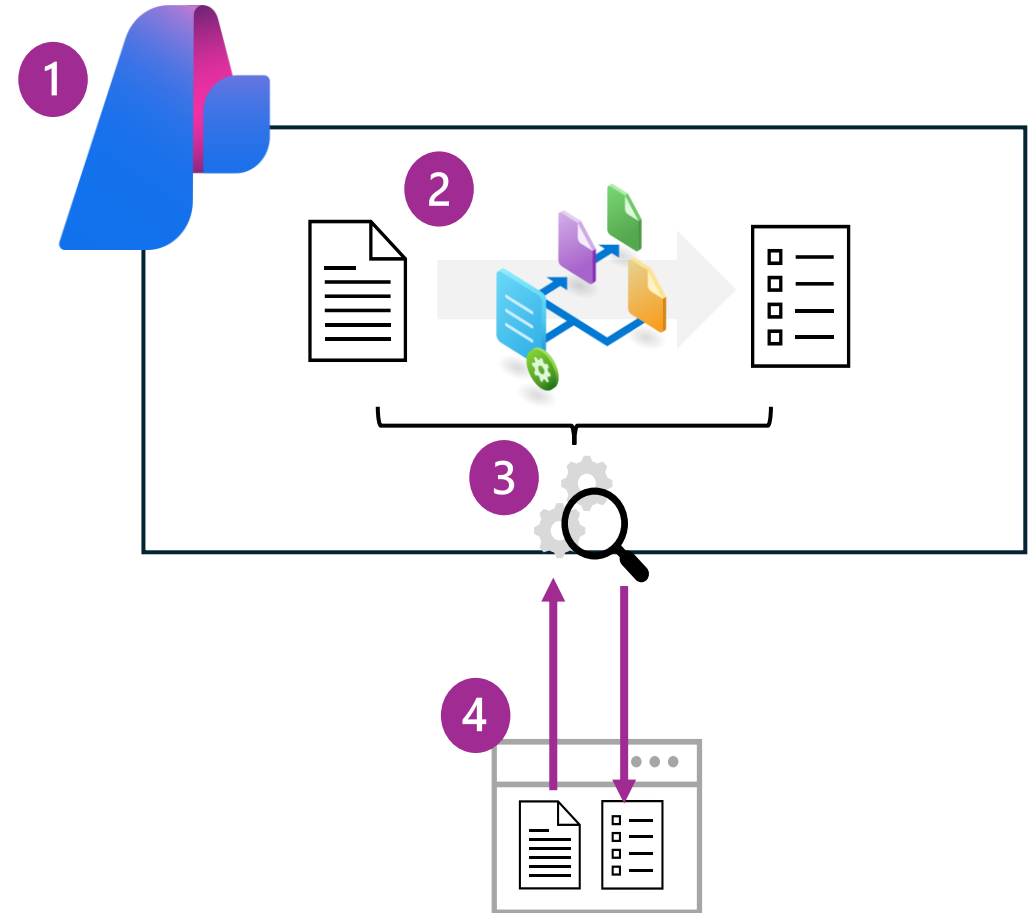
[Check token usage in JSON \(Result tab\) and learn about pricing here](#)

☒ Your privacy choices

Privacy & cookies Terms of use © Microsoft 2026

Creating an analyzer

1. Create a Microsoft Foundry project
2. Define a Content Understanding schema for the information to be extracted
 - This can be based on a content sample and an analyzer template
3. Build an analyzer based on the completed schema
4. Use the analyzer to extract or generate fields from new content



Exercise

Extract information from multimodal content

In this exercise, you'll use Azure Content Understanding to:

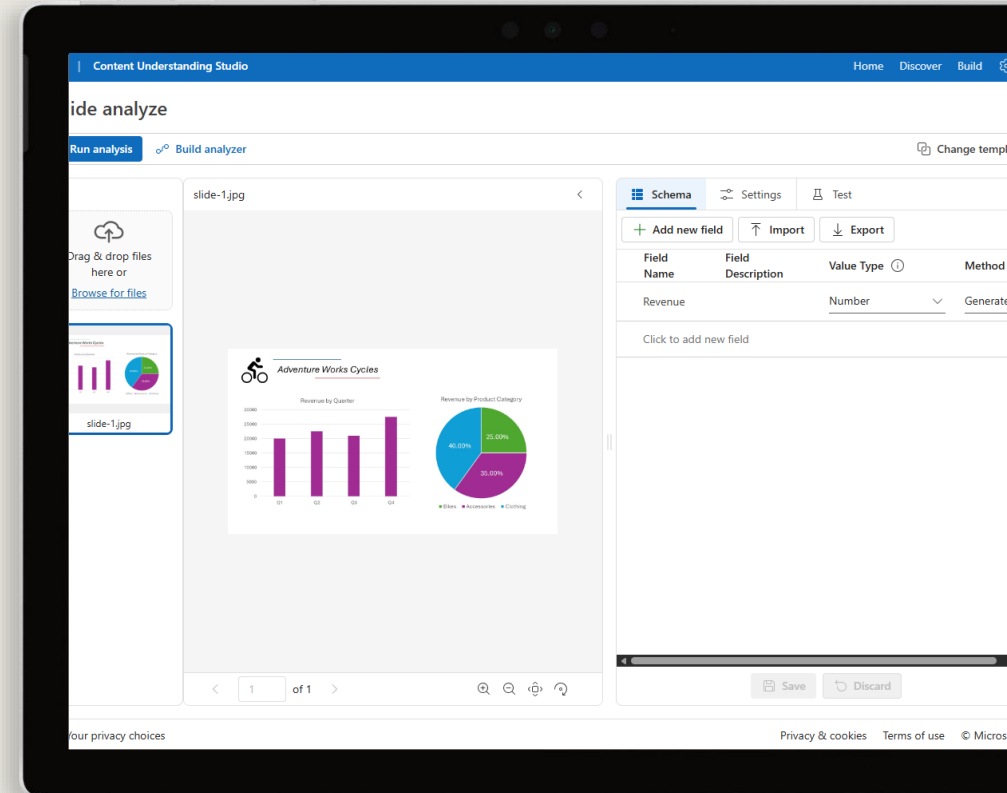
- Analyze documents
- Analyze slides
- Analyze audio
- Analyze video

Start the exercise at:

21. Extract information from multimodal content

🕒 1 h

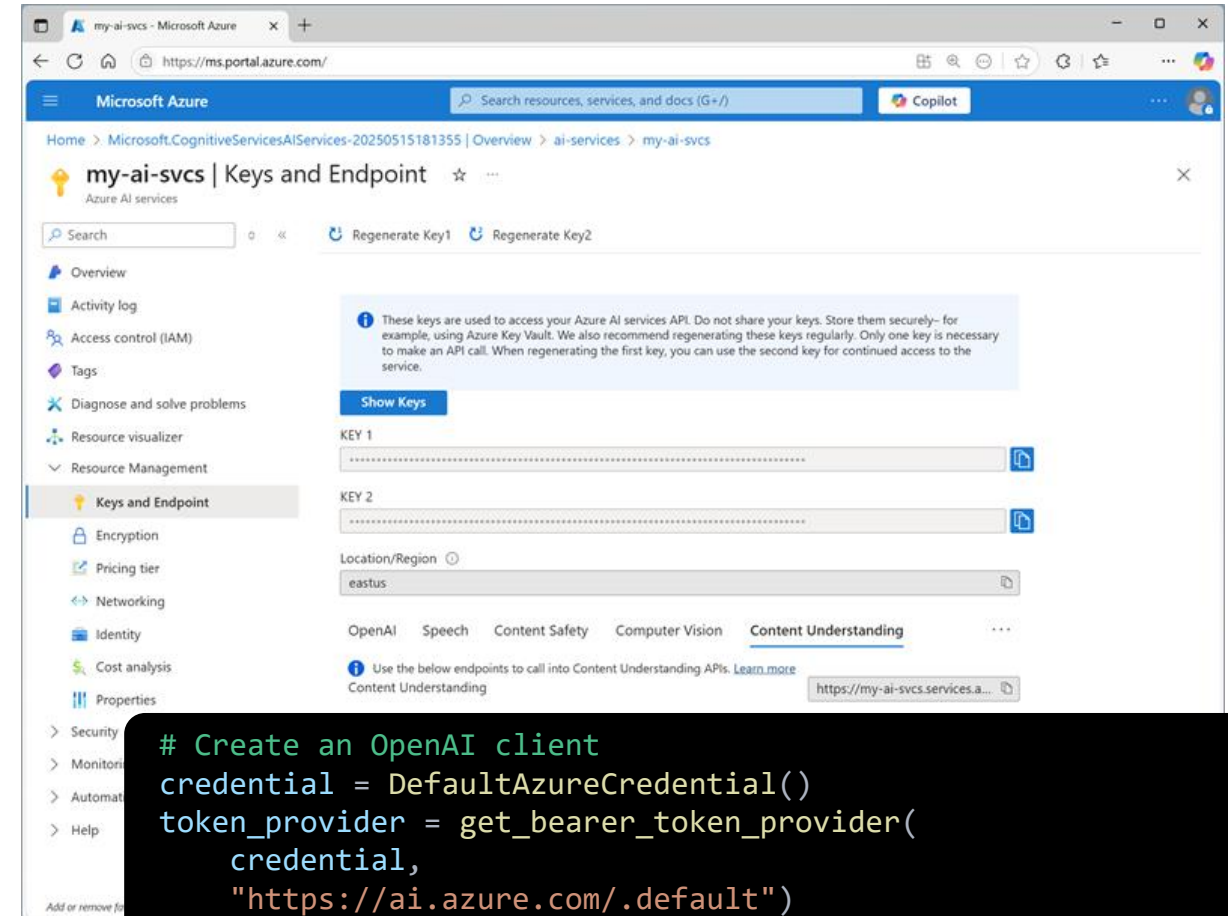
0%



Prepare to use the Content Understanding API

To use the Azure Content Understanding API:

- Create a Microsoft Foundry project.
- Authenticate via Entra ID or Content Understanding endpoint and API key
 - To find your credentials via code: use the Microsoft Foundry SDK with Microsoft Entra ID to automatically retrieve connection details from your project.



```
# Create an OpenAI client
credential = DefaultAzureCredential()
token_provider = get_bearer_token_provider(
    credential,
    "https://ai.azure.com/.default")

client = OpenAI(
    base_url=openai_endpoint,
    api_key=token_provider()
)
```

Define your schema and create analyzer

- Define the fields you want to extract or generate
- Specify the base analyzer and generative models to use
- Send schema to client to create analyzer

```
client.begin_create_analyzer(  
    analyzer_name,  
    body=analyzer_definition)
```

- Key components:
 - `baseAnalyzerId` – Base analyzer (e.g., `prebuilt-document`)
 - `fieldSchema` – Fields with type, method (extract/generate), and description
 - `models` – Completion and embedding models to use

```
{  
  "description": "Simple business card",  
  "baseAnalyzerId": "prebuilt-document",  
  "config": {  
    "returnDetails": true  
  },  
  "fieldSchema": {  
    "fields": {  
      "ContactName": {  
        "type": "string",  
        "method": "extract",  
        "description": "Name on business card"  
      },  
      "EmailAddress": {  
        "type": "string",  
        "method": "extract",  
        "description": "Email address on business card"  
      }  
    }  
  },  
  "models": {  
    "completion": "gpt-4.1",  
    "embedding": "text-embedding-3-large"  
  }  
}
```

Analyze content

1. Authenticate to your client
2. Begin analysis, providing:
 - analyzer_name
 - input (URL or binary data)
3. Retrieve the result
4. Extract contents

John Smith

Email: john@contoso.com

```
"contents": [
  {
    "markdown": "John Smith\nEmail: john@contoso.com\n",
    "fields": {
      "ContactName": {
        "type": "string",
        "valueString": "John Smith",
        "spans": [
          {
            "offset": 0,
            "length": 10
          }
        ],
        "confidence": 0.994,
        ...
      },
      "EmailAddress": {
        "type": "string",
        "valueString": "john@contoso.com",
        "spans": [
          {
            "offset": 18,
            "length": 16
          }
        ],
        "confidence": 0.998,
        ...
      }
    }
  },
  ...
]
```

Exercise

Develop a Content Understanding client application

In this exercise, you'll use the Content Understanding API to:

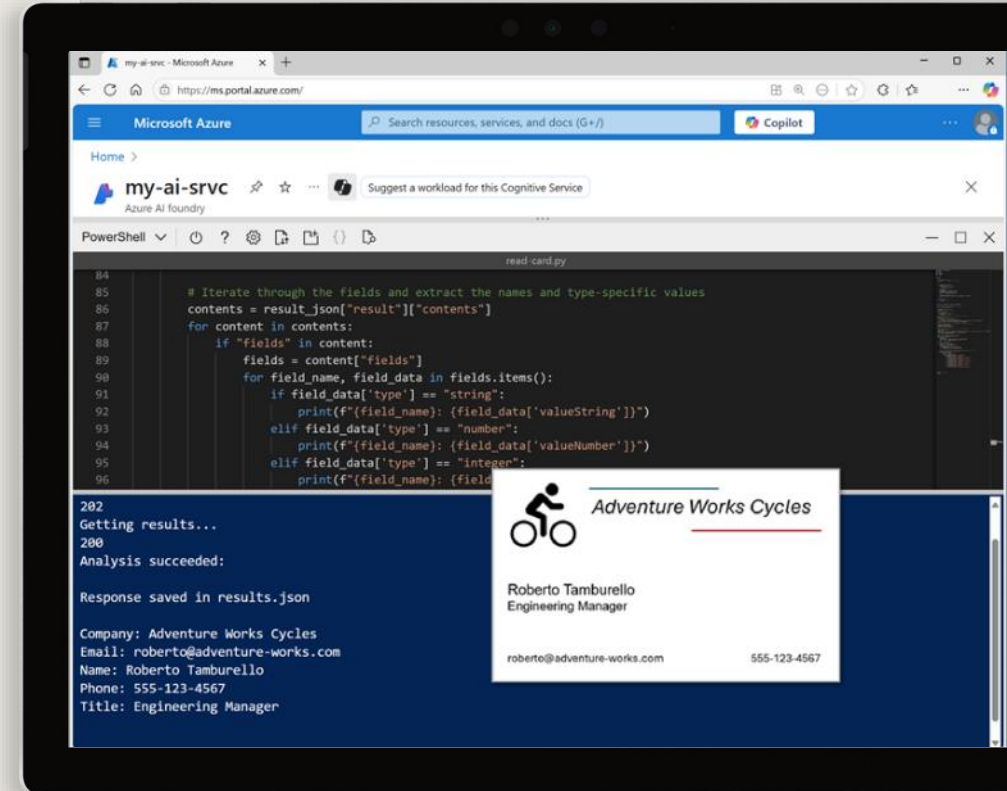
- Create an analyzer
- Analyze content

Start the exercise at:

22. Develop a Content Understanding client application

🕒 1 h

0%



Knowledge check



1 What kinds of AI solutions is Azure Content Understanding designed to help you build?

- ☐ Chatbots that automatically translate between multiple spoken and written languages
- ☒ Analyzers that extract information from documents, images, videos, and audio files
- ☐ Image generators that create visualizations based on descriptions

2 Which graphical tool should you use to create an Azure Content Understanding project?

- ☐ Microsoft Visual Studio
- ☐ Azure Machine Learning studio
- ☒ Content Understanding studio

3 What should you define for the information you want to extract from content?

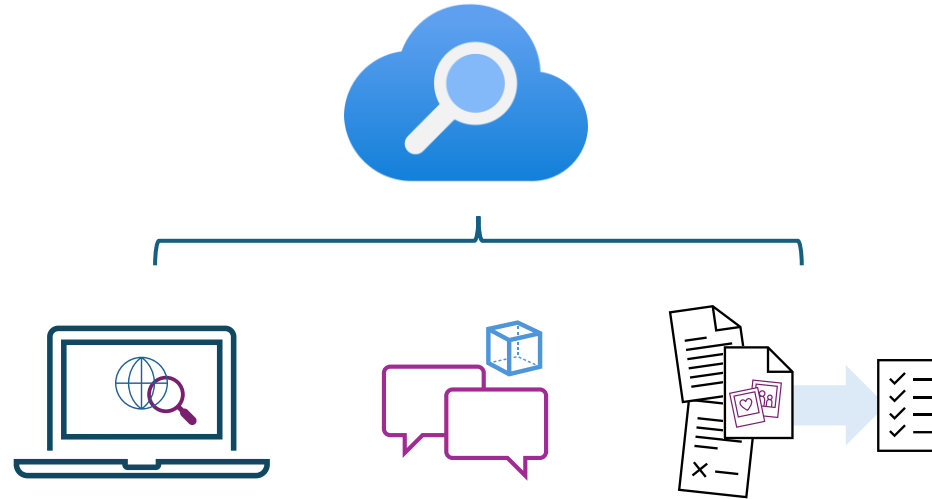
- ☒ A schema
- ☐ An index
- ☐ A cluster

Create a knowledge mining solution with Azure AI Search

<https://aka.ms/mslearn-ai-knowledge-mining>



What is Azure AI Search?



Key capabilities:

- *Intelligent indexing*: Index documents and data from a range of sources, including images
- *AI enrichment*: Use AI skills to enrich index data
- *Vectorization*: Support agentic workloads

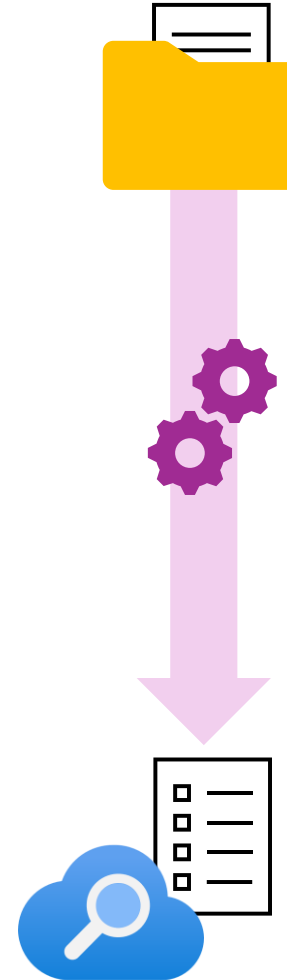
Core applications:

- *Enterprise Search*: Powering websites and apps to find information fast.
- *RAG for Generative AI*: Grounding LLM prompts
- *Knowledge Mining*: Turning massive document libraries into data assets for analytics.*

*Our focus today

Extract data with an indexer

- An *indexer* extracts data from a *data source*
- *Document cracking* retrieves the text content from the source
- An *enrichment pipeline* iteratively builds JSON representations of each indexed document
- The *fields* for each document can be:
 - Extracted from the source (e.g. file name, date, size)
 - Inferred or generated in the enrichment pipeline
- The result is an *index* containing all of the indexed documents



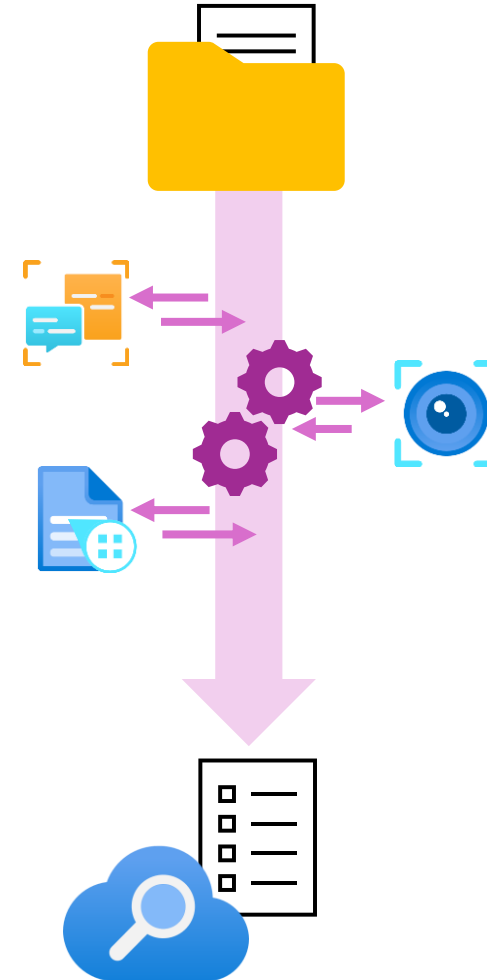
Enrich extracted data with AI skills

Built-in skills:

- Detecting the language that text is written in
- Detecting and extracting places, locations, and other entities in the text
- Determining and extracting key phrases within a body of text
- Translating text
- Identifying and extracting (or removing) personally identifiable information (PII) within the text
- Extracting text from images
- Generating captions and tags to describe images
- Multimodal image and text vectorization

Custom skills:

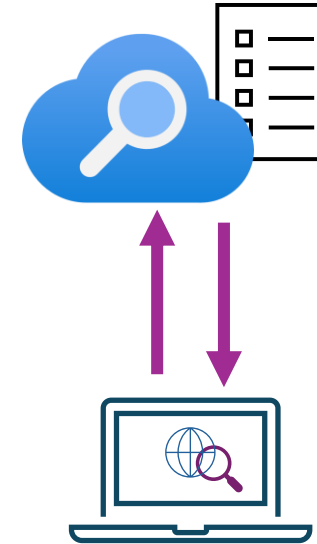
- Implement custom logic to return fields based on input
- Often “wrappers” around services, like Azure Language



Search an index

Each index field can be configured as:

- **key**: Fields that define a unique key for index records
- **searchable**: Fields that can be queried using full-text search
- **filterable**: Fields that can be included in filter expressions to return only documents that match specified constraints
- **sortable**: Fields that can be used to order the results
- **facettable**: Fields that can be used to determine values for *facets* (user interface elements used to filter the results based on a list of known field values)
- **retrievable**: Fields that can be included in search results

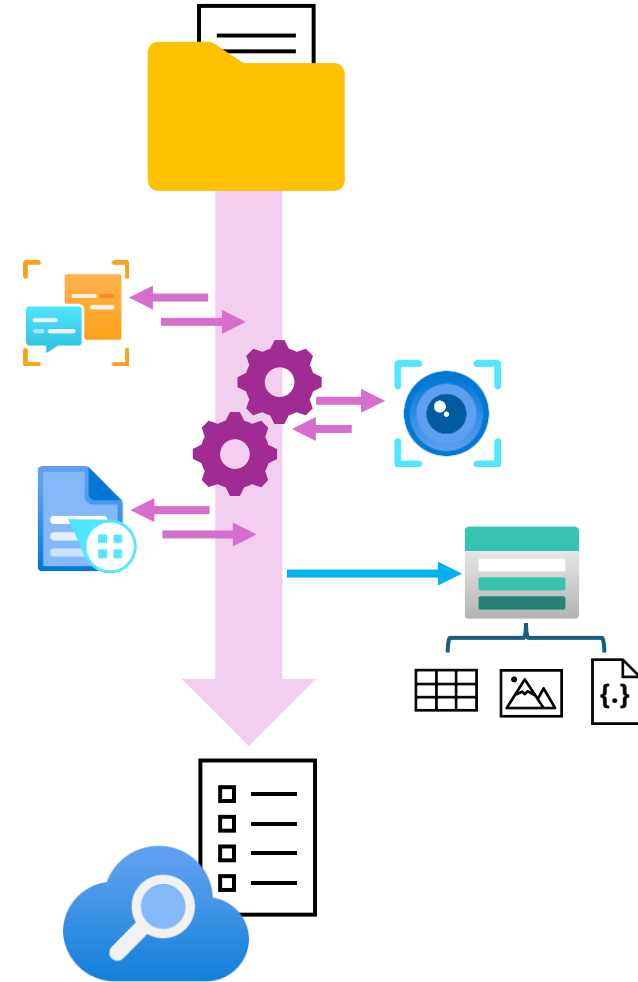


```
{  
  "search": "New York",  
  "select": "title,locations,keyphrases",  
  "filter": "author eq 'Reviewer'",  
  "orderby": "metadata_storage_date",  
}
```

Persist extracted information in a knowledge store

Store *projections* in Azure Storage:

- **Objects:** JSON documents
- **Tables:** Relational schema for extracted fields
- **Files:** Extracted images



Demo

Create a knowledge mining solution

In this exercise, you'll use Azure AI Search to:

- Create an index
- Query an index
- Persist projections in a knowledge store

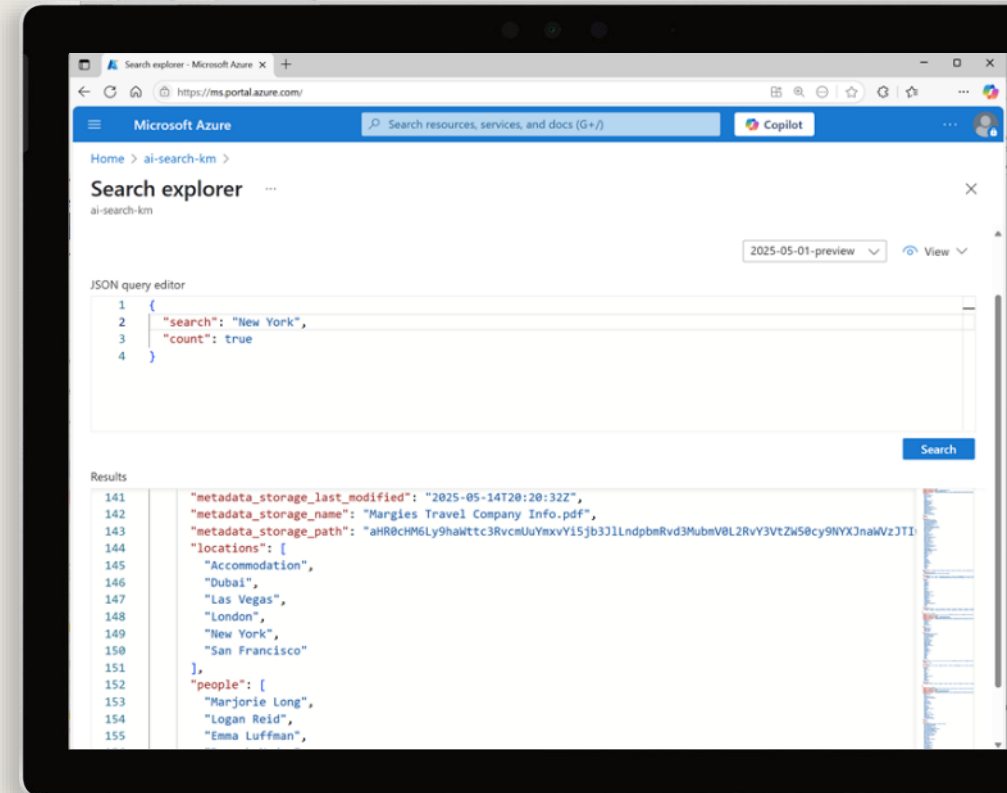
Start the exercise at:

23. Create a knowledge mining solution



1 h

0%



Knowledge check



1 Which component of an Azure AI Search solution is scheduled to extract and enrich data to populate an index?

- ☒ Indexer
- ☐ Projection
- ☐ Query

2 Which service supports built-in AI skills in Azure AI Search?

- ☐ Azure Functions
- ☒ Foundry Tools
- ☐ Azure Cosmos DB

3 Which kind of projection results in a relational data schema for extracted fields?

- ☐ File
- ☐ Object
- ☒ Table

